

RAILSYNC.AI

Automatic Block Planning, Cross-Departmental Coordination & Optimization System Indian Railways • Ministry of Railways (CRIS / RDSO Architecture)

COMPLETE PROJECT DOSSIER, FULL APPLICATION PAGE-BY-PAGE SPECIFICATION, R&D; RESEARCH LINKS & JUDGES' PITCH
GUIDE

Version 2.4 (Production Grade) • Confidential & Academic Documentation

1. Executive Summary & Problem Landscape

Indian Railways operates 13,000+ passenger trains and 8,000+ freight rakes daily. Maintenance is currently scheduled in manual departmental silos (Civil Track/TMS, Signalling/SMMS, Traction/TDMS, and Control Office/COA). Because departments operate independently, the same track section is closed 3 to 4 times a week, causing cascading train delays, freight demurrage penalties, and maintenance backlogs.

2. Multi-Disciplinary Shadow Block Concept

RAILSYNC.AI synchronizes compatible cross-departmental maintenance tasks under unified 'Shadow Blocks' fitted precisely into off-peak train headway gaps. This cuts total track closure hours by 47.8% while preserving full safety buffers.

Metric	Manual Siloed Planning	RAILSYNC.AI Shadow Block
Possession Structure	3 separate uncoordinated blocks	1 unified shadow block
Total Track Downtime	115–135 mins / week	60 mins / week (47.8% saved)
Passenger Train Delay	3–5 train path disruptions	0 conflicts (Zero delay)
Corridor Availability	88.5% capacity	95.8% capacity (+7.3%)
Re-Planning Speed	3–4 hours manual paper flow	< 120 ms (OR-Tools CP-SAT)

3. Complete Application Page-by-Page Deep Dive (All Modules)

Comprehensive architectural breakdown of all 20 pages and subsystems in the RAILSYNC.AI application:

Page 1: Executive Operations Dashboard (ExecutiveDashboard.tsx)

- **What it Consists of:** • Live Golden Diagonal Route Topology (NDLS–PRYJ via TDL) with moving train telemetry pulse.
- 8 Core Operations KPIs: Total Fixed Assets (1,845), Demurrage Saved (₹48.2L), Critical Defects (19), Pending Tasks (142), Available Blocks (24), Asset Availability (96.4%), Hours Saved (34.8%), Conflicts Avoided (14).
- 7-Day Rolling Corridor Asset Availability Trajectory comparing manual baseline (88.5%) vs AI-optimized (95.8%).
- Weekly Maintenance Block Utilization chart (hours allocated vs utilized, 94.2% average efficiency).
- 3 Target Breakdown Panels: Tasks by Department (TMS/SMMS/TDMS), Tasks by RDSO Priority Scale, and Train Headway Safety Index (82% zero-conflict buffer).
- **How it Works:** • Pulls real-time summary statistics from the FastAPI backend endpoint ``/api/dashboard/summary`` backed by PostgreSQL 18.
- Displays dynamic AI operations recommendations computed by the OR-Tools solver.
- All 8 KPI tiles and network sections are interactive, allowing 1-click drill-down navigation into underlying registries and planning modules.

Page 2: Corridor & Network Topology (RailwayNetwork.tsx)

- **What it Consists of:** • Detailed sectional mapping of the Golden Diagonal Corridor across 6 sectors (SEC-01 New Delhi-Ghaziabad, SEC-02 Ghaziabad-Aligarh, SEC-03 Aligarh-Tundla, SEC-04 Tundla Chord, SEC-05 Tundla-Etawah, SEC-06 Etawah-Kanpur).
- Track specifications: Quad Track, Double Line, Electrification (25kV AC), Section Speed (130/160 km/h), Maximum Axle Load (25t), TQI Health Index.
- **How it Works:** • Fetches structured spatial data from ``/api/sections``.
- Allows engineers and dispatchers to inspect sectional bottleneck indices, temporary speed restrictions (TSR), and scheduled shadow blocks across individual railway segments.

Page 3: Train Timetables & COA Feed (TrainSchedule.tsx)

- **What it Consists of:** • Integrated Control Office Application (COA) live timetable feed.
- Tracks passenger expresses (Vande Bharat 22436, Rajdhani 12424, Shatabdi 12004, Gomti Express 12420) and heavy freight rakes (BOXN Coal, BCN Fertilizer).
- Arrival, departure, directional transit paths, sectional speeds, and headway safety windows.

- **How it Works:** • Ingests timetable records from ``/api/trains`` and ``/api/trains/schedules``.
- Identifies clean operational gaps (e.g. 02:00–04:30 IST night curfew) where multi-disciplinary maintenance can be performed without delaying high-priority passenger services.

Page 4: Available Block Windows & Matrix (PlanningCalendar.tsx)

- **What it Consists of:** • Registry of certified engineering possession slots granted by the Operating Department.
- Shows date, start/end time, available duration, allowed department limit (up to 3 parallel disciplines), and status (Available, Reserved, Approved).
- **How it Works:** • Connects to ``/api/blocks``.
- Acts as candidate time-slot inputs for the Google OR-Tools CP-SAT combinatorial scheduling engine.

Page 5: Cross-Department Live Telemetry Stream (DepartmentIntegration.tsx)

- **What it Consists of:** • Unified real-time telemetry hub bridging Track Maintenance System (TMS), Signal & Telecom (SMMS), Traction/OHE (TDMS), and Control Office (COA).
- TMS Feed: Ultrasonic Flaw Detection (USFD) crack logs, Track Recording Car (TRC) geometry runs, TQI scores.
- SMMS Feed: Point Machine operating current telemetry (perm. $\leq 4.2A$), Digital Axle Counter wheel counts, Interlocking status.
- TDMS Feed: 25 kV AC catenary tension (kgf), contact wire residual thickness (mm), spark-gap calibration, isolator resistance.
- COA Feed: Active train paths, headway safety restrictions, power permit coordination.
- **How it Works:** • Pulls consolidated telemetry from ``/api/departments/integration-data`` with auto-sync polling every 5 seconds.
- Harmonizes siloed departmental data into unified JSON schema feeds for the AI Compatibility & Scheduling Engine.

Page 6: Loco Pilot Telemetry Feedback (LocoPilotFeedback.tsx)

- **What it Consists of:** • Crowdsourced operational feedback submitted by Loco Pilots and Assistant Loco Pilots (V-Crew).
- Captures track jerk/vibration, rail abnormal sound, signal visibility obstruction, and OHE pantograph sparking with GPS kilometer markers.
- Verification workflow with Section Engineer sign-off and automatic AI urgency booster (+15-25 points).
- **How it Works:** • Connects to ``/api/loco-pilot/feedback``.
- Verified loco pilot complaints automatically spawn high-priority maintenance work orders and trigger dynamic re-planning.

Page 7: Maintenance Tasks Repository (MaintenanceTasks.tsx)

- **What it Consists of:** • Master repository of all pending, scheduled, and completed maintenance work orders across Track, S&T, and Traction.
- Displays Task Code, Department, Section, AI Urgency Score, Estimated Duration, and Loco Pilot linkage.
- 1-Click CSV Export and 'AI Auto-Prioritize Queue' button.
- **How it Works:** • Connects to ``/api/tasks`` and ``/api/ai/prioritize-all``.
- Allows filtering by priority level (Critical, High, Medium, Low), department, and corridor section.

Page 8: Fixed Assets Registry (AssetsPage.tsx)

- **What it Consists of:** • Comprehensive inventory of fixed railway infrastructure assets: Turnouts, Rail Segments (CWR/LWR), Point Machines, Digital Axle Counters, Contact Wires, Cantilevers, and TSS Isolators.
- Location (km), health condition score (0-100), criticality grade, and operational status.
- **How it Works:** • Connects to ``/api/assets``.
- Maps asset degradation curves to inform preventive maintenance scheduling before failures occur.

Page 9: Defect & Flaw Detection Registry (DefectsPage.tsx)

- **What it Consists of:** • Real-time safety defect registry capturing Ultrasonic Flaw Detection (USFD) transverse fatigue cracks, Alumino-thermic (AT) weld flaws, point machine stroke backlash, and contact wire wear.
- Severity classification (IMR - Immediate Removal, REM - Early Removal, OBS - Observe).
- **How it Works:** • Connects to ``/api/defects``.
- Feeds directly into the ML Priority Engine to enforce mandatory emergency block allocation.

Page 10: AI Priority Center - ML Urgency Scoring (AIPriorityCenter.tsx)

- **What it Consists of:** • Interactive Machine Learning evaluation sandbox running a trained RandomForest Regressor.
- Parameter Sliders: Asset Criticality, Defect Severity, Urgency Score, Overdue Days, Asset Condition, Safety Risk, Operational Impact, Loco Pilot Observations.
- Live Radar Chart of 7 Operational Dimensions, ML Prediction Gauge (0-100), and Explainable AI (XAI) feature importance breakdown.
- **How it Works:** • Ingests parameters and calls ``/api/ai/prioritize`` to compute mathematical urgency scores and plain-English factor justifications.

Page 11: AI Automatic Block Planner - OR-Tools CP-SAT (OptimizationCenter.tsx)

- **What it Consists of:** • Core automated multi-disciplinary block optimizer.
- Section selector (SEC-01 to SEC-04) and 'GENERATE OPTIMIZED BLOCK PLAN' action.
- Parallel Gantt Chart Timeline: Visualizes Track, S&T, and Traction tasks executing concurrently under 1 traffic block.
- Before vs After Comparison: Proves 47.8% track hours saved, 0 express clashes, and +7.3% availability gain.
- Alternative Candidate Windows scoring table (Top 3 candidate slots).
- 'WHY THIS BLOCK?' Explainable AI Modal and 'ADD UNEXPECTED FREIGHT' scenario trigger.
- Human-in-the-Loop Governance: APPROVE, MODIFY (with start/end time sliders), and REJECT controls.
- **How it Works:** • Submits combinatorial optimization requests to `/api/optimization/run``.
- Google OR-Tools CP-SAT evaluates candidate blocks, enforces 15-min passenger headway buffers, and returns optimal shadow blocks in <120 ms.

Page 12: Weekly Plan Matrix - 7-Day Tactical Roster (PlanningCalendar.tsx - Weekly)

- **What it Consists of:** • Tactical 7-day calendar matrix (Monday to Sunday).
- Day-by-day block cards showing Block ID, Section, Window Time, Departments bundled, Task counts, Conflict Risk, and Approval status.
- Interactive Day Selection showing detailed possession schedule and 'Open in Block Optimizer' action.
- **How it Works:** • Synchronizes with top navbar 'Weekly Plan' toggle and renders tactical schedules with 1-click export to CSV/PDF.

Page 13: Monthly Plan Matrix - 4-Week Strategic Roadmap (PlanningCalendar.tsx - Monthly)

- **What it Consists of:** • Strategic 4-week corridor roadmap (Week 1 to Week 4).
- Weekly summary cards: Total Tasks Planned, Critical Overdue Cleared, Planned Blocks, and Availability Trajectory (94.2% -> 97.4%).
- Strategic capacity synergy callouts.
- **How it Works:** • Synchronizes with top navbar 'Monthly Plan' toggle to provide senior management with rolling 30-day capacity forecasts.

Page 14: Operations Simulation Engine (Simulation.tsx)

- **What it Consists of:** • Discrete-event traffic simulator comparing 'Manual Sequential Baseline' vs 'AI-Optimized Shadow Plan'.
- Train Path String Chart / Time-Distance Diagram.
- Conflict detection matrix calculating headway clashes, speed restriction penalties, and net passenger punctuality.
- **How it Works:** • Calls `/api/simulation/run`` to simulate train paths minute-by-minute across the corridor.

Page 15: What-If Scenario Analysis - Digital Twin (WhatIfAnalysis.tsx)

- **What it Consists of:** • Real-time perturbation sandbox testing corridor resilience against 3 disturbance presets:
 1. Unexpected Freight Influx (NTPC Dadri Coal Rake BCN-429 inserted into 02:10-02:50 slot).
 2. Emergency IMR Rail Defect (Ultrasonic crack at Turnout 204 requiring urgent preemption).
 3. Resource / Crew Diverted (S&T; gang missing, dynamically unbundling into a 45-min dual block).
- Interactive Sliders: Freight Delay Offset (0-120 mins), TSR Speed Limit (15-110 km/h), Emergency Preemption Toggle.
- 3-Way Comparative Diff: Original Baseline vs Unmitigated Impact vs AI Re-Optimized Solution.
- 'Submit Re-Plan for Approval' action transmitting re-routes to the Chief Controller.
- **How it Works:** • Sends perturbations to `/api/what-if/run`` and re-solves the mathematical model in under 120 milliseconds.

Page 16: Human Approvals Governance (Approvals.tsx)

- **What it Consists of:** • Official electronic sign-off queue for Divisional Railway Officers.
- Displays pending block proposals, departments coordinated, safety impact assessment, and automated explanation summary.
- Digital Approve, Reject, and Request Modification actions with auditable officer comments.
- **How it Works:** • Connects to `/api/approvals``, upholding the governance principle that AI proposes while authorized officers sign off.

Page 17: System Alerts & Operational Notifications (Notifications.tsx)

- **What it Consists of:** • Live operational alert stream across TMS, SMMS, TDMS, and Control Office dispatchers.
- Filter by All, Critical Only (IMR flaw alerts), and Unread.
- 1-Click 'Mark All as Read' action.
- **How it Works:** • Connects to `/api/notifications`` and reflects unread alert count badge ('4') in the top navbar.

Page 18: Operational Efficiency & Audit Reports (Reports.tsx)

- **What it Consists of:** • Comprehensive audit dashboard for RDSO and Railway Board inspections.
- Metric views across Weekly, Monthly, and Quarterly periods: Total Blocks Executed, Block Hours Utilized, Possession Hours Saved (47.4%), Passenger Punctuality (98.2%), Demurrage Avoided (■14.8L), Corridor Availability Score (96.4%).
- Departmental bundling breakdown and verified audit compliance logs.
- 1-Click Downloads for Complete Project Dossier (PDF), Complete Dossier (Word DOCX), and Audit CSV.
- **How it Works:** • Connects to `/api/reports/summary`` and `/api/docs/download/{file_type}``.

Page 19: System Settings & Configuration (Settings.tsx)

- **What it Consists of:** • Engine parameter tuning: Headway buffer threshold (15 mins), Optimization objective weights, solver timeout (1000 ms), auto-sync intervals.
- Database status and connected subsystem health.
- **How it Works:** • Allows authorized administrators to calibrate optimization weights and connection strings.

Page 20: Role-Based Access Control & Login (Navbar & Login.tsx)

- **What it Consists of:** • Top navbar Multi-Persona Switcher supporting 5 Railway Personas:
 1. Shri Rajeshwar Sharma, IRTS (Chief Operations Manager - CPTM, HQ Level 14)
 2. Smt. Ananya Sen, IRSE (Senior Divisional Engineer - Sr. DEN / P-Way)
 3. Shri Vikramaditya Rao, IRSSE (Senior Divisional Signal & Telecom Engineer - Sr. DSTE)
 4. Shri Harpreet Singh, IRSEE (Senior Divisional Electrical Engineer - Sr. DEE / Traction)
 5. Shri Manoj Deshmukh (Chief Controller & Power Controller - Operating Department)
- Custom JWT credentials login page.
- **How it Works:** • Persists user state in `localStorage` and transmits JWT Bearer tokens with full RBAC permission enforcement.

4. Mathematical Optimization Formulation (Google OR-Tools CP-SAT)

Objective Function: Maximize $Z = \sum(w_priority * Priority_i * X_i,j) + \sum(w_shadow * Synergy_Bonus_j) - \sum(w_train * Clash_Penalty_j) - \sum(w_duration * Duration_j)$

Hard Constraints: Mandatory 15-minute passenger train headway safety buffer, machine resource exclusivity (tamping machines & tower wagons), and 25kV traction power isolation lockstep.

5. Machine Learning Urgency Scoring & Explainable AI (XAI)

RandomForest Regressor ingests USFD flaw detection logs, TRC geometry car indices (TQI), asset criticality, and real-time Loco Pilot feedback. Generates auditable, plain-English rationales explaining why each block was selected and how conflicts were avoided.

6. Official R&D; References & Research Links

1. **RDSO Track Maintenance Manual (2024):** Technical specs for USFD, TRC & Track Geometry (<https://rdso.indianrailways.gov.in>)
2. **Indian Railway Permanent Way Manual (IRPWM):** Traffic Block Regulations (<https://indianrailways.gov.in>)
3. **CRIS Control Office Application (COA):** Timetable & Freight Dispatch System (<https://cris.org.in>)
4. **Google OR-Tools CP-SAT:** Combinatorial Transportation Optimization (<https://developers.google.com/optimization>)

7. Pitch Guide: 3-Minute Presentation to Judges

[0:00–0:45] The Hook: Highlight the daily challenge of 21,000+ trains and 3–4 uncoordinated closures on the same track.

[0:45–1:30] The Solution: Introduce RAILSINC.AI and the Coordinated Multi-Disciplinary Shadow Block innovation.

[1:30–2:30] Live Demo: Show ML Priority ranking, click 'OPTIMIZE' (118 ms solve time, 47.8% hours saved), and trigger What-If freight influx.

[2:30–3:00] Governance: Emphasize officer-in-the-loop approval: 'AI predicts, optimization schedules, authorized officers approve.'

8. Top Judges' Q&A;

Q: How is train safety guaranteed?

A: The CP-SAT solver strictly enforces mandatory 15-minute headway buffers before and after passenger trains. Overlapping blocks are mathematically prohibited.

Q: Is human authority bypassed?

A: No. AI only generates optimal recommendations; final execution requires authorized digital sign-off via RBAC.